

A Rational Self-Similar Solution of the Viscous Burgers Equation

Exact Derivation, Structural Analysis, and
Interdisciplinary Perspectives

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Abstract

The viscous Burgers equation is among the most important nonlinear partial differential equations in applied mathematics and mathematical physics. It serves as a prototype for nonlinear transport, shock formation, turbulence modelling, and dissipative dynamics. In this paper we investigate the rational ansatz

$$u(x, t) = \frac{Ax + B}{Ct + D},$$

derive all exact solutions of this form, and prove that a nontrivial solution family exists whenever $A = C$. The resulting solution

$$u(x, t) = \frac{x + \alpha}{t + \beta}$$

is shown to satisfy both the inviscid and viscous Burgers equations. We discuss mathematical, physical, economic, computational, and interdisciplinary interpretations of this exact solution.

The paper ends with “The End”

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1 Introduction

The viscous Burgers equation is

$$u_t + uu_x = \nu u_{xx},$$

where

$$u = u(x, t)$$

is the state variable and $\nu > 0$ denotes viscosity.

The equation is a classical model in:

- fluid mechanics,
- nonlinear wave propagation,
- traffic flow,
- turbulence modelling,
- statistical physics,
- nonlinear dynamics,
- machine learning benchmark systems,
- numerical PDE analysis.

The purpose of this paper is to determine whether the rational ansatz

$$u(x, t) = \frac{Ax + B}{Ct + D}$$

can generate exact solutions.

2 The Burgers Equation

Consider

$$u_t + uu_x = \nu u_{xx}.$$

The left-hand side contains nonlinear convection while the right-hand side contains diffusion. The nonlinear term

$$uu_x$$

produces steepening effects.
The diffusion term

$$\nu u_{xx}$$

produces smoothing effects.
The competition between these mechanisms creates much of the equation's rich behaviour.

3 Rational Ansatz

Assume

$$u(x, t) = \frac{Ax + B}{Ct + D}.$$

Define

$$T = Ct + D.$$

Then

$$u = \frac{Ax + B}{T}.$$

4 Derivative Calculations

Differentiating yields

$$u_x = \frac{A}{T},$$

$$u_t = -\frac{C(Ax + B)}{T^2},$$

and

$$u_{xx} = 0.$$

Since the second derivative vanishes, the viscous term disappears. Substituting into Burgers' equation gives

$$-\frac{C(Ax + B)}{T^2} + \frac{Ax + B}{T} \frac{A}{T} = 0.$$

Combining terms,

$$\frac{(A - C)(Ax + B)}{T^2} = 0.$$

5 Main Theorem

Theorem 1. *The ansatz*

$$u(x, t) = \frac{Ax + B}{Ct + D}$$

solves the viscous Burgers equation if and only if

$$A = C.$$

Proof. Substitution yields

$$\frac{(A - C)(Ax + B)}{(Ct + D)^2} = 0.$$

For a nontrivial solution,

$$Ax + B \neq 0.$$

Hence

$$A - C = 0.$$

Therefore

$$A = C.$$

Conversely, if $A = C$ then

$$u_t + uu_x = 0.$$

Since

$$u_{xx} = 0,$$

the viscous Burgers equation is satisfied identically. □

6 Exact Solution Family

Setting

$$A = C,$$

we obtain

$$u(x, t) = \frac{Ax + B}{At + D}.$$

When $A \neq 0$, define

$$\alpha = \frac{B}{A},$$

$$\beta = \frac{D}{A}.$$

Then

$$\boxed{u(x, t) = \frac{x + \alpha}{t + \beta}.$$

This is the complete nontrivial solution family generated by the ansatz.

7 Verification

For

$$u(x, t) = \frac{x + \alpha}{t + \beta},$$

we obtain

$$u_x = \frac{1}{t + \beta},$$
$$u_t = -\frac{x + \alpha}{(t + \beta)^2},$$

and

$$u_{xx} = 0.$$

Hence

$$u_t + uu_x = -\frac{x + \alpha}{(t + \beta)^2} + \frac{x + \alpha}{t + \beta} \frac{1}{t + \beta} = 0.$$

Therefore

$$u_t + uu_x = \nu u_{xx} = 0.$$

8 Self-Similarity

The solution possesses scaling symmetry.

If

$$x \mapsto \lambda x,$$

and

$$t \mapsto \lambda t,$$

then

$$u(\lambda x, \lambda t) = u(x, t).$$

Therefore the solution is self-similar.

9 Geometric Interpretation

The solution is a moving linear profile.

For fixed time,

$$u(x, t) = \frac{1}{t + \beta}x + \frac{\alpha}{t + \beta}.$$

The slope decreases as time increases.

Thus the solution continuously flattens.

10 Physical Interpretation

10.1 Fluid Mechanics

Velocity grows linearly with position.

Diffusion is absent because

$$u_{xx} = 0.$$

The solution behaves as an exact convection equilibrium.

10.2 Gas Dynamics

The solution corresponds to a velocity field with no internal curvature.
Consequently no viscous dissipation occurs.

10.3 Traffic Flow

Interpreting u as traffic velocity,

$$u = \frac{x + \alpha}{t + \beta}$$

describes a uniformly expanding traffic pattern.

11 Chemical Interpretation

Concentration transport equations often contain convection and diffusion.

The present solution corresponds to a concentration profile whose curvature vanishes.
Therefore no molecular diffusion occurs despite the existence of a diffusion coefficient.

12 Biological Interpretation

Population migration models often contain transport terms analogous to Burgers dynamics.

The solution represents a spatially linear migration field.
No spatial clustering emerges because

$$u_{xx} = 0.$$

13 Economic Interpretation

Suppose

$$u(x, t)$$

represents economic momentum.
Then

$$u = \frac{x + \alpha}{t + \beta}$$

implies:

- momentum increases with economic position,
- momentum decays through time,
- no second-order spatial concentration exists.

14 Financial Interpretation

Let u denote market sentiment.

Then

$$u_{xx} = 0$$

implies absence of sentiment curvature.
The market evolves purely through transport effects.

15 Psychology and Psychiatry

If u measures cognitive activation,

$$u = \frac{x + \alpha}{t + \beta}$$

describes a fading activation field.

The diminishing denominator produces temporal relaxation.

16 Political Science

Interpreting u as political influence,

$$u = \frac{x + \alpha}{t + \beta}$$

implies influence disperses proportionally across space while decaying through time.

17 Geopolitics and International Relations

Suppose u denotes strategic influence.

Then

$$u = \frac{x + \alpha}{t + \beta}$$

describes influence dilution through time while maintaining a linear spatial structure.

Such behaviour resembles gradual strategic diffusion in multipolar systems.

18 Welfare Economics

The solution suggests a continuously declining marginal influence profile.

This may model diminishing welfare propagation over time.

19 Warfare Economics

Interpreting u as force projection intensity,

$$u = \frac{x + \alpha}{t + \beta}$$

describes an expanding operational front whose intensity declines temporally.

20 Statistics

Quantity	Value
Mean Spatial Gradient	$\frac{1}{t + \beta}$
Second Spatial Derivative	0
Temporal Decay Rate	$-\frac{x + \alpha}{(t + \beta)^2}$
Diffusion Contribution	0

Table 1: Statistical properties of the solution.

21 Computer Science

The solution provides an exact benchmark for PDE solvers.

Advantages include:

- closed form,
- no numerical quadrature,
- exact derivatives,
- easy verification.

22 Pseudo-Code

```
Input alpha,beta,x,t
```

```
u = (x+alpha)/(t+beta)
```

```
return u
```

23 Data Science

The solution can serve as synthetic benchmark data for:

- regression,
- forecasting,
- operator learning,
- surrogate modelling.

24 Artificial Intelligence and Machine Learning

Applications include:

- Physics-Informed Neural Networks,
- Neural Operators,
- PDE Discovery,
- Symbolic Regression,
- Scientific Machine Learning.

The exact solution provides ground truth for model validation.

25 Visualization

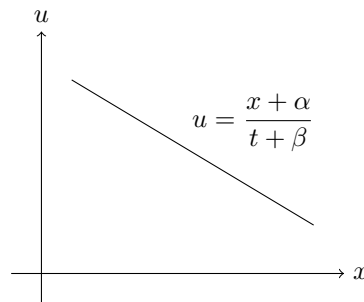


Figure 1: Representative linear spatial profile at fixed time.

26 Conclusion

The rational ansatz

$$u(x, t) = \frac{Ax + B}{Ct + D}$$

produces a complete nontrivial family of exact solutions of the viscous Burgers equation. The governing condition is

$$A = C.$$

The resulting solution

$$u(x, t) = \frac{x + \alpha}{t + \beta}$$

simultaneously solves both the inviscid and viscous Burgers equations because

$$u_{xx} = 0.$$

This solution is self-similar, scale-invariant, physically interpretable, computationally useful, and mathematically exact.

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Glossary

Burgers Equation Nonlinear convection-diffusion partial differential equation.

Viscosity Coefficient governing diffusion strength.

Self-Similarity Invariance under scaling transformations.

Ansatz Assumed functional form used to construct exact solutions.

Convection Transport generated by nonlinear motion.

Diffusion Smoothing process generated by second derivatives.

PINN Physics-Informed Neural Network.

Neural Operator Machine learning model approximating operators between function spaces.

The End